

ASHAB ALI KHAN

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Professional Summary

- AI Engineer specializing in **LLMs, RAG, Arabic NLP, and Multi-Agent Systems** with hands-on experience building and deploying production AI applications. Developed an **Arabic-English RAG Chatbot** for bilingual PDF document querying and a **6-Agent GCC Business Intelligence Platform** that automates Gulf market research, report generation, and email delivery — both live on Hugging Face. Proficient in **Python, LangChain, LangGraph, ChromaDB, Sentence Transformers, Prompt Engineering, and Semantic Search**. Available for immediate relocation to **UAE/KSA**.

Technical Skills

- AI & Machine Learning:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Agentic AI, LangChain, LangGraph, Scikit-learn, TensorFlow, Keras.
- Programming & Data:** Python, C++, SQL, NumPy, Pandas.
- NLP & AI Tools:** NLP, Arabic NLP, Embeddings, Sentence Transformers, Semantic Search, Vector Search, ChromaDB, Groq API, Tavily, Tesseract OCR.
- Tools & Deployment:** Docker, Streamlit, Hugging Face Spaces, Git, GitHub, Google Colab, REST APIs, API Integration.
- Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Software Engineering.

Projects

- GCC AI & Startup Intelligence Platform ([LIVE](#))** Personal Project
Jun 2026
[GitHub Repository](#)
 - Architected **6-agent LangGraph pipeline** processing **30+ web sources** across **6 GCC countries**, delivering professional Word reports via email in under **2 minutes**.
 - Built **dynamic Query Understanding Agent** supporting **6 query types** — startup discovery, funding intelligence, comparisons, sector reports, hiring, and investor research.
 - Resolved **LLM token limit errors** by implementing **batch processing** of 6 sources per LLM call, enabling full 30+ source pipeline without crashes.
 - Developed **Review Agent** with keep/reject decision per entity, filtering **13–18 irrelevant items** per query before report generation to improve output quality.
 - Integrated **Tavily, Groq LLM, SendGrid**, and **python-docx** into single automated pipeline with **Streamlit** dashboard and **Docker** deployment.
- Arabic RAG Chatbot ([LIVE](#))** Personal Project
Jan 2026
[GitHub Repository](#)
 - Built a **bilingual Arabic-English RAG chatbot** for Arabic PDF querying using **LangChain, ChromaDB, and Arabic-Triplet-Matryoshka-V2 embeddings** (768-dim, top-3 retrieval).
 - Diagnosed **non-standard font encoding corruption** in PDF extraction; switched to **Tesseract OCR**, extracting **81 pages** reliably across all tested Arabic documents.
 - Resolved **cross-lingual retrieval failure** by translating English queries to Arabic before embedding, aligning both into the same vector space and improving semantic retrieval accuracy.
 - Engineered **hybrid conversation memory** — last 2 full exchanges plus LLM-generated rolling summary — keeping token cost **constant across unlimited conversation length**.
 - Deployed on **Hugging Face Spaces** via Docker; diagnosed and fixed **403 upload errors** caused by Streamlit XSRF misreading reverse proxy headers through custom CORS configuration.

Education

- Galgotias University** Greater Noida, UP
2023 - 2025
Master of Computer Application (MCA)
- Mahatma Jyotiba Phule Rohilkhand University (MJPRU)** Bareilly, UP
2019 - 2023
Bachelor of Science (BSc)